



Assessment Analysis Guide
26-27 Math G4 Form D Factors/Multiples
Math
Grade 4

Reference #: A1751f052

1

Item Name: i185072

Item Type: SR (Selected Response)

Standards Description:

4.OA.B.4 Find all factor pairs for a whole number in the range 1–100. Recognize that a whole number is a multiple of each of its factors. Determine whether a given whole number in the range 1–100 is a multiple of a given one-digit number. Determine whether a given whole number in the range 1–100 is prime or composite.

Content:

Which pair of numbers are both factors of 60?

- A 5 and 36
- B 9 and 30
- C 12 and 30
- D 25 and 35

Primary Standard: 4.OA.B.4**Points Possible:** 1**Distractor Rationale:**

- | | |
|---------------------|--|
| A: 5 and 36 | Student selected a factor of the given number and a number that shares factors with the given number but is not itself a factor. |
| B: 9 and 30 | Student identified a multiple of one of the factors rather than the factor itself. |
| C: 12 and 30 | Correct. |
| D: 25 and 35 | Student identified numbers based on their sum rather than their product, OR selected numbers that share a common factor with the given number. |

2

Item Name: i185094

Item Type: SR (Selected Response)

Standards Description:

4.NF.A.2 Compare two fractions with different numerators and different denominators, e.g., by creating common denominators or numerators, or by comparing to a benchmark fraction such as $\frac{1}{2}$. Recognize that comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with symbols $>$, $=$, or $<$, and justify the conclusions, e.g., by using a visual fraction model.

Content:

Which of the following makes the comparison true?

$\frac{1}{2} < \underline{\hspace{1cm}}$

- A

$\frac{1}{3}$
- B

$\frac{3}{12}$
- C

$\frac{5}{10}$
- D

$\frac{6}{8}$

Primary Standard: 4.NF.A.2

Points Possible: 1

Distractor Rationale:

- A:

$\frac{1}{3}$

Student may have thought that when fractions have the same numerator, the fraction with the greater denominator is larger, not recognizing that a larger denominator actually results in a smaller fraction.
- B:

$\frac{3}{12}$

Student may have thought that a greater numerator and denominator always result in a greater fraction, not recognizing that the overall value of a fraction depends on the relationship between the numerator and denominator.
- C:

$\frac{5}{10}$

Student did not recognize that the fractions were equivalent and may have thought that a greater numerator and denominator always result in a greater fraction, not recognizing that the overall value of a fraction depends on the relationship between the numerator and denominator.
- D:

$\frac{6}{8}$

Correct.

3

Item Name: i185037

Item Type: SR (Selected Response)

Standards Description:

- 4.OA.B.4
- Find all factor pairs for a whole number in the range 1–100. Recognize that a whole number is a multiple of each of its factors. Determine whether a given whole number in the range 1–100 is a multiple of a given one-digit number. Determine whether a given whole number in the range 1–100 is prime or composite.

Content:

Which number is a multiple of both 4 and 6?

- A

2
- B

8
- C

12
- D

18

Primary Standard: 4.OA.B.4

Points Possible: 1

Distractor Rationale:

- A:

2

Student confused a common factor with a common multiple.
- B:

8

Student chose a number that is a multiple of only one given number when looking for a common multiple between two numbers.
- C:

12

Correct.
- D:

18

Student chose a number that is a multiple of only one given number when looking for a common multiple between two numbers.

4

Item Name: i185038

Item Type: MS (Multiple Select)

Standards Description:

- 4.OA.B.4

Find all factor pairs for a whole number in the range 1–100. Recognize that a whole number is a multiple of each of its factors. Determine whether a given whole number in the range 1–100 is a multiple of a given one-digit number. Determine whether a given whole number in the range 1–100 is prime or composite.

Content:

Select **three** numbers that are prime.

- A 2
- B 11
- C 15
- D 18
- E 43

Primary Standard: 4.OA.B.4

Points Possible: 1

Distractor Rationale:

- | | | |
|----|----|--|
| A: | 2 | Correct. |
| B: | 11 | Correct. |
| C: | 15 | Student possibly thought all odd numbers are prime, OR confused prime and composite. |
| D: | 18 | Student possibly confused prime and composite. |
| E: | 43 | Correct. |